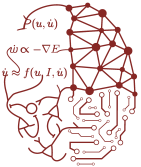
# Asynchronous Integrated Circuits
## An Introduction




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Thursday 18th December, 2025

UNIVERSITÄT BERN

# Part I - INTEGRATED CIRCUITS

GUT meeting 06/01/2026

## INTEGRATED CIRCUITS

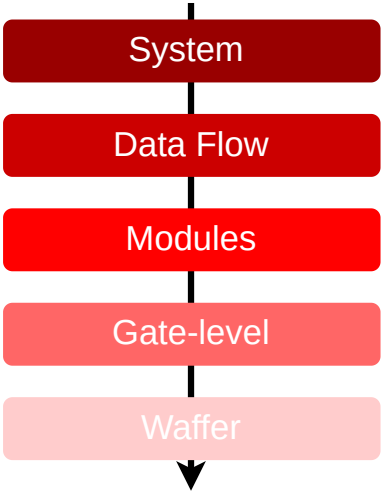
1. Abstractions
   - Layers
   - Combinational Logic
2. Design Families
3. Design Flow

## ASYNCHRONOUS CIRCUITS

4. Linear pipeline or digital Fordism
5. Clock-driven
6. Event-driven
7. Bundled-Data

# Layers

## Abstractions




System on Chip, IPs, Memories, IOs

Pipeline, Algorithm

Functionnal blocs, Procedures

Logic gates, Flip-Flops, Latches, Standard Cells

Mask

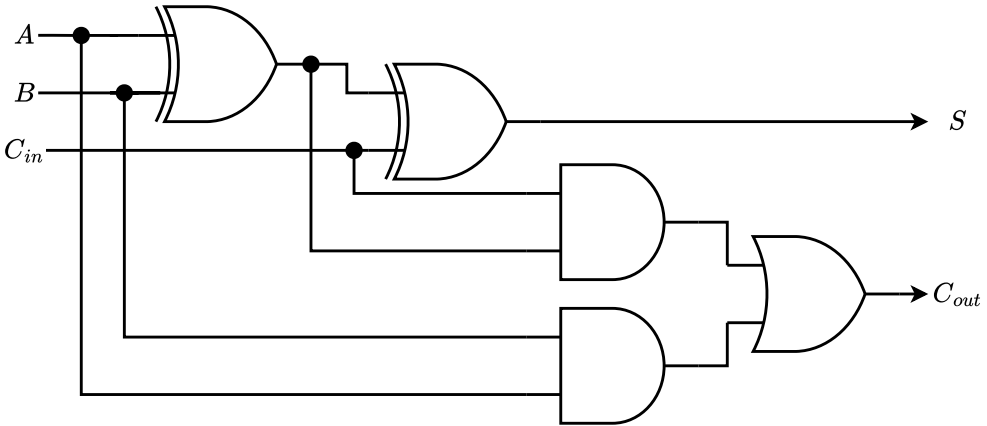
$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in} \quad (1)$$

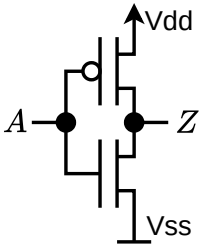
$$S = A \oplus B \oplus C_{in} \quad (2)$$

$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in} \quad (3)$$

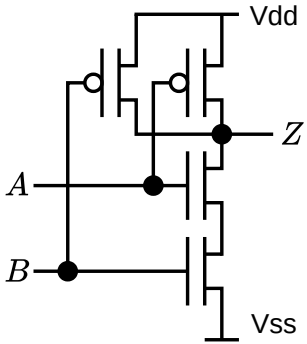
$$C_{out} = AB + C_{in}(A \oplus B) \quad (4)$$

| Input | | | Output | |
|---|---|---|---|---|
| $A$ | $B$ | $C_{in}$ | $S$ | $C_{out}$ |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

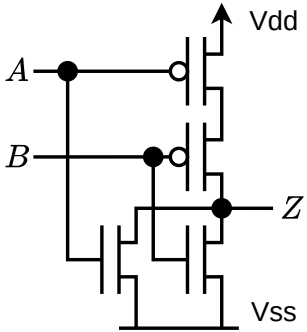

A
B
$C_{in}$
S
$C_{out}$



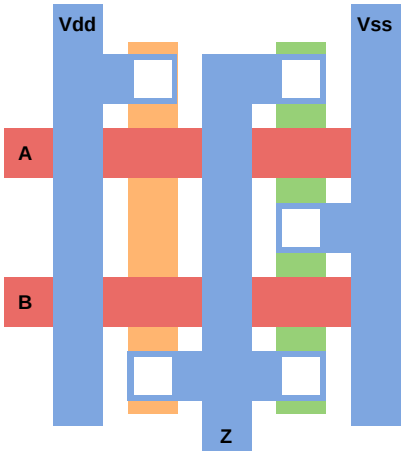

Inverter gate CMOS




NAND gate CMOS




NOR gate CMOS




Metal

Gate (polysilicon)

P diffusion

N diffusion

**INTEGRATED CIRCUITS**

1 Abstractions

2 Design Families
- Overview
- Central Processing Unit
- Graphical Processing Unit
- Field-Programmable Gate Array [7]
- System on Chip

3 Design Flow

**ASYNCHRONOUS CIRCUITS**

4 Linear pipeline or digital Fordism

5 Clock-driven

6 Event-driven

7 Bundled-Data

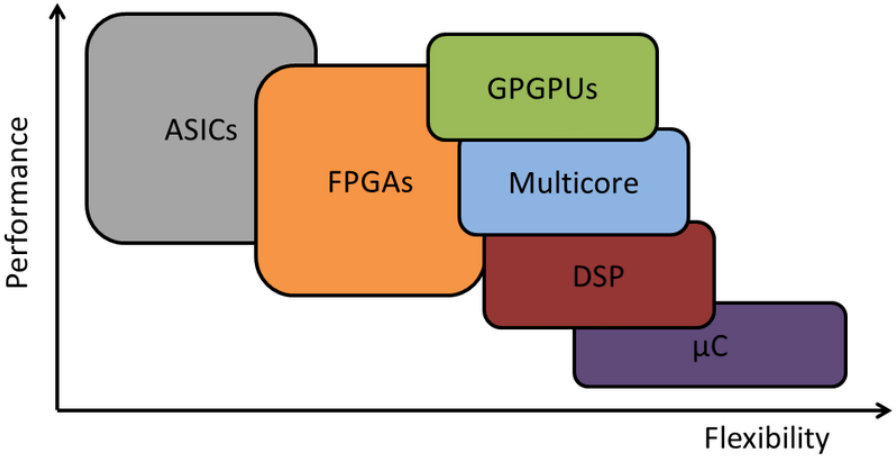



Figure: Performance vs Flexibility [1]

- Instruction Set Architecture
- Single Instruction Single Data (SISD)
- Executes a program

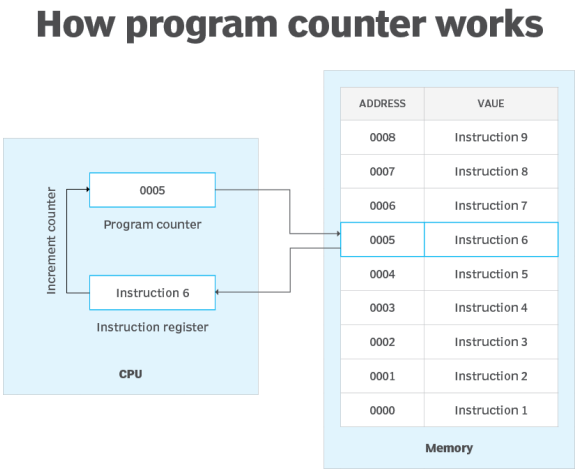



Figure: Program [2]

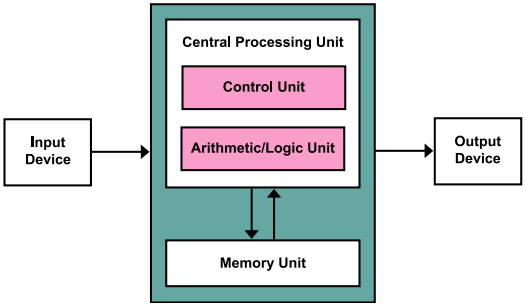



Figure: Von Neumann Architecture [3]

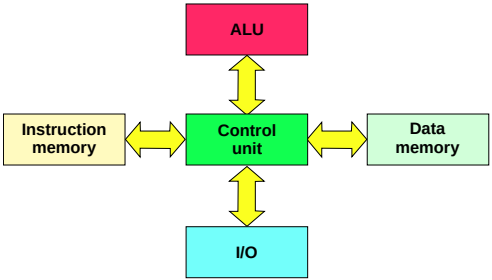



Figure: Harvard Architecture [4]

- IF: Instruction Fetch
- ID: Instruction Decode
- EX: Execute
- MEM: Memory access
- WB: WriteBack

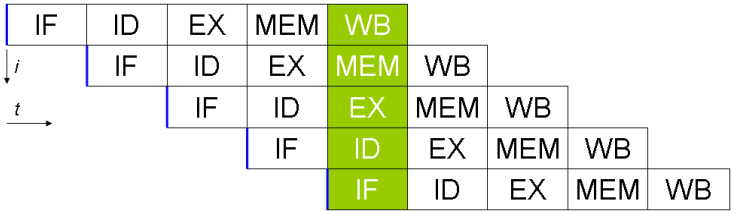



Figure: Five stage pipeline [5]

- Single Instruction Multiple Data (SIMD)
- Vector operations
  - scalar multiplication $a\vec{v}$

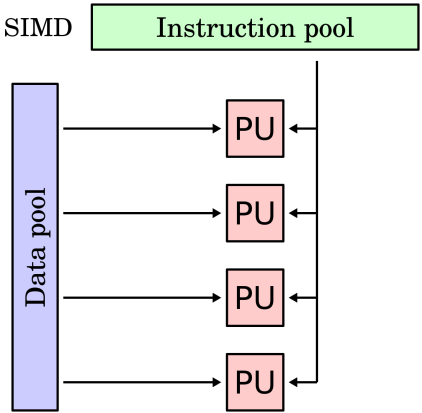
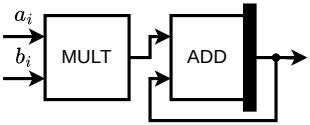


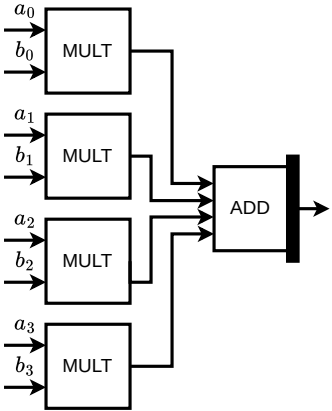

Figure: [6]

Multiply ACcumulation (MAC) operation in dot product $\vec{a} \cdot \vec{b}$

Serial:

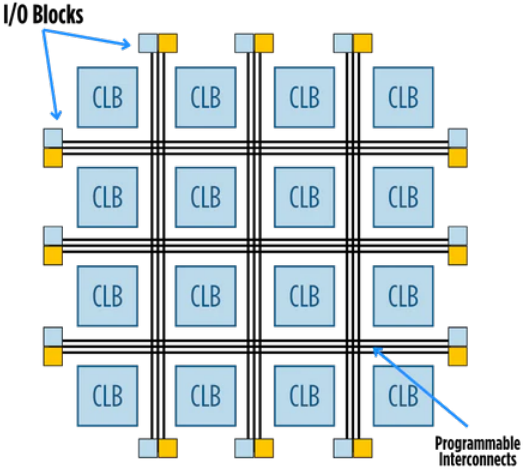


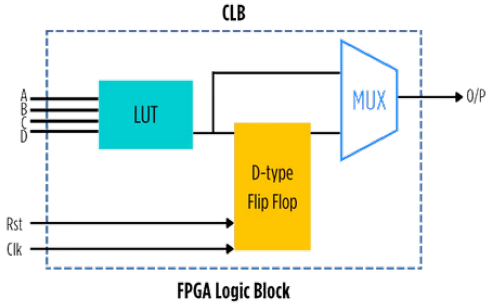

Parallel:

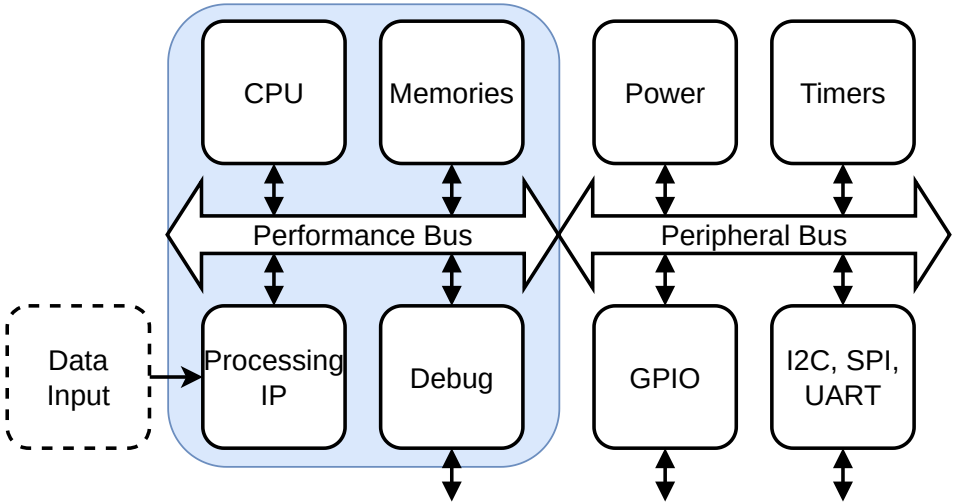



**Comparison**

- Size
- Power consumption
- Latency


I/O Blocks
CLB
Programmable
Interconnects


CLB
A
B
C
D
LUT
MUX
O/P
D-type
Flip Flop
Rst
Clk
FPGA Logic Block


CPU
Memories
Power
Timers
Performance Bus
Peripheral Bus
Data Input
Processing IP
Debug
GPIO
I2C, SPI, UART

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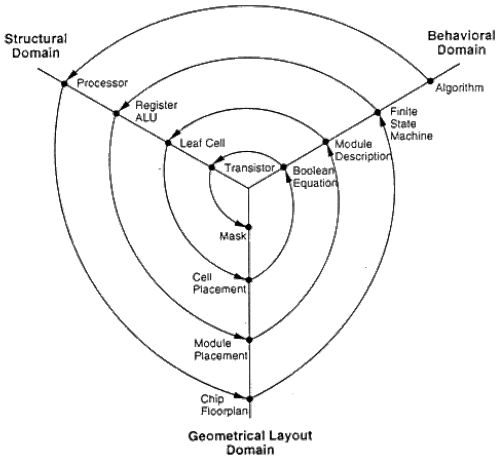
Design Flow


**INTEGRATED CIRCUITS**

1. Abstractions
2. Design Families
3. Design Flow
   - Successive Refinement
   - Verification & Validation

**ASYNCHRONOUS CIRCUITS**

4. Linear pipeline or digital Fordism
5. Clock-driven
6. Event-driven
7. Bundled-Data




Three perspectives:

- Behavioral
- Structural
- Physical/Geometry

**Logical synthesis** : from behavioral to structural

**Physical synthesis / Layout** : from structural to physical

**High Level Synthesis**

- From system level description
- Produces a Finite State Machine (FSM) (RTL level description)
- Followed by logical synthesis

**Logical synthesis**

- Takes a RTL description & technology node
- Builds a abstract representation of the logic and maps it to standard cell
- Produces a netlist / gate-level circuit

**Steps**

- Partitioning
- Floorplanning
- Placement
- Global routing
- Detailed routing

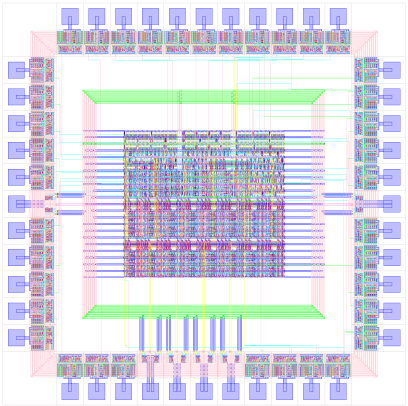


Figure: Chip layout [8]

## Verification

- Functionnal Simulation
  - Behavioral
  - Gate-level (Timed)
  - Analog
- Formal Verification

## Validation

- FPGA (Behavioral)
- Static Timing Analysis and Timing Closure

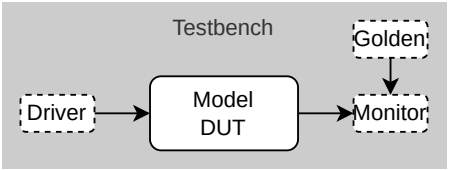



Figure: Simulated Testbench

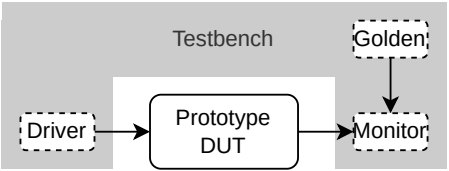



Figure: Hardware in the loop

# Summary

- The IC design flow consist of successive refinement accross abstraction layers
- Trade-offs : Flexibility vs Performance, Serial vs Parallel, etc.
- CPUs and GPUs are instruction set architectures that execute programs
- System on chip are composed of multiple subsystems (IPs)
- FPGA are a good platform to prototype hardware designs

# Part II - ASYNCHRONOUS CIRCUITS

GUT meeting 13/01/2026

## INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

## ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism
- Concepts

5 Clock-driven

6 Event-driven

7 Bundled-Data

# Assembly Line

## Linear pipeline or digital Fordism

# Linear Pipeline

## Linear pipeline or digital Fordism

# Latency & Throughput

## Linear pipeline or digital Fordism - Concepts

## INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

## ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism

5 Clock-driven
- Timing Constraints
- Drawbacks

6 Event-driven

7 Bundled-Data

# Clock Signal

## Clock-driven

# Linear Pipeline

## Clock-driven - Timing Constraints

# Hold

## Clock-driven - Timing Constraints

# Clock Skew

## Clock-driven - Drawbacks

## INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

## ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism

5 Clock-driven

6 Event-driven
- [TODO] Communication
- Transactions at the gate level
- Formalism

7 Bundled-Data

# Local Synchronization

## Event-driven - [TODO] Communication

# Handshakes

## Event-driven - [TODO] Communication

# Delay Insensitive Circuits

## Event-driven - Transactions at the gate level

# State-Holding gates

## Event-driven - Transactions at the gate level

# Signal Transition Graph

## Event-driven - Formalism

**INTEGRATED CIRCUITS**

1 Abstractions

2 Design Families

3 Design Flow

**ASYNCHRONOUS CIRCUITS**

4 Linear pipeline or digital Fordism

5 Clock-driven

6 Event-driven

7 Bundled-Data
- Timing Constraints

# Transaction at the pipeline level

## Bundled-Data -

# Setup & Hold

## Bundled-Data - Timing Constraints

# Summary

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10 Metastability & Arbiters

11 Properties of Petri Nets & STG

12 Families of Bundled-Data circuits

# References I


[1] R. Molanes, K. Amarasinghe, J. Rodriguez-Andina, and M. Manic, "Deep learning and reconfigurable platforms in the internet of things: Challenges and opportunities in algorithms and hardware," *IEEE Industrial Electronics Magazine*, vol. 12, pp. 36–49, Jun. 2018. DOI: 10.1109/MIE.2018.2824843

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# Problematic

## Oligopoly problem in VLSI

# OPENROAD

## Oligopoly problem in VLSI

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# Metastability in Arbiters

## Metastability & Arbiters

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# Occupancy & Static Spread

## Families of Bundled-Data circuits

# Capture

## Families of Bundled-Data circuits

# Latched Controller

## Families of Bundled-Data circuits